

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

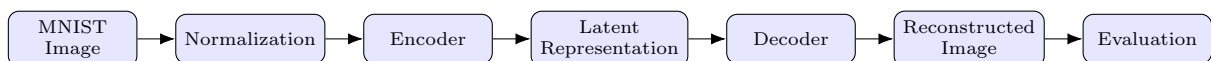
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

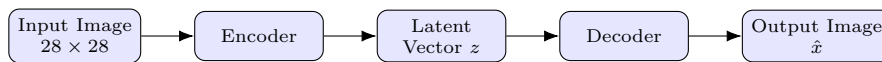
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

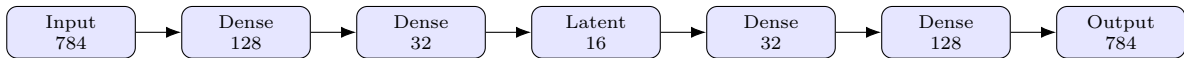
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

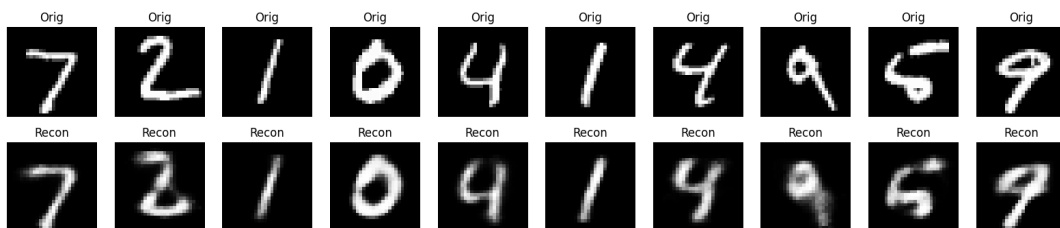


Figure 1: Original versus Reconstructed Images

Plot 2: Training and Validation Reconstruction Loss

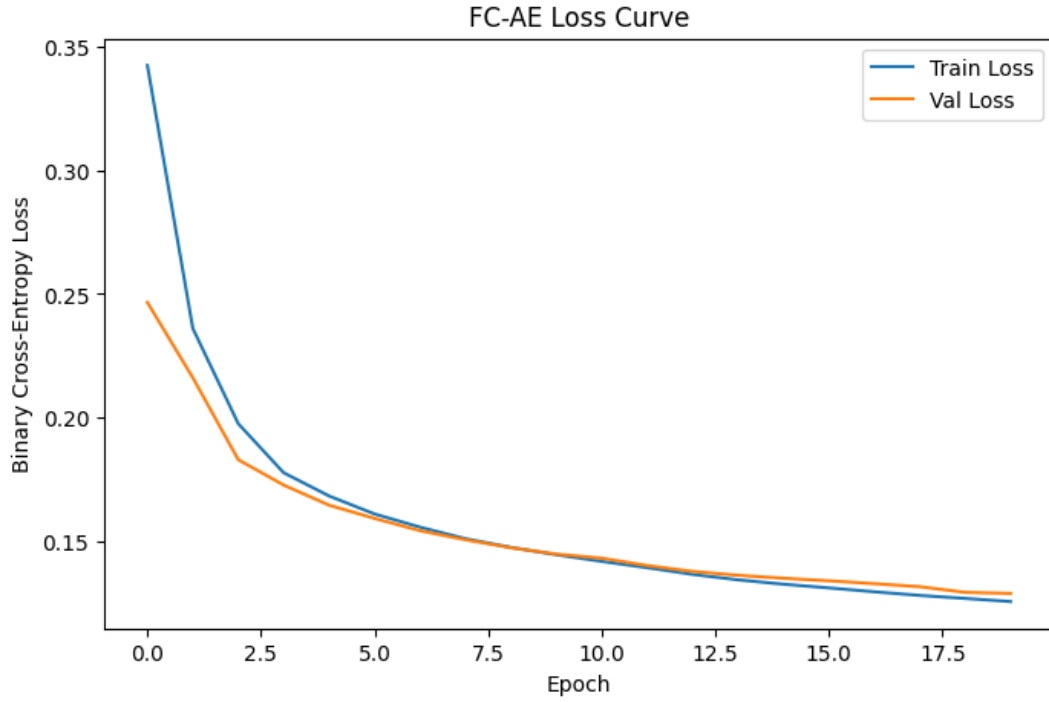


Figure 2: Training and Validation Reconstruction Loss

Inference: The loss and validation loss graphs decline sharply during the initial epochs and then flatten out, demonstrating that the model converges well and without any overfitting since both curves lie very close to each other.

9. Reconstruction Metrics

Calculated the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

SSIM between the original and reconstructed images.

The implementation uses:

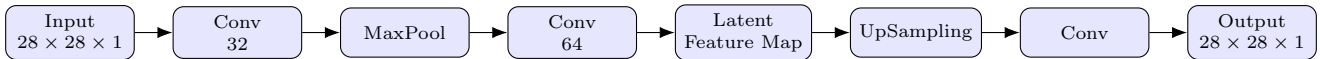
<https://scikit-image.org/docs/stable/api/skimetrics.html>

Metric	Value
Test MSE	0.022776
Test MAE	0.059414
Mean SSIM	0.729679

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



The implemented architecture uses the Keras functional API:

```

conv_input = layers.Input(shape=(28, 28, 1))
x = layers.Conv2D(32, 3, activation='relu', padding='same')(conv_input)
x = layers.MaxPooling2D(2, padding='same')(x)
x = layers.Conv2D(64, 3, activation='relu', padding='same')(x)
x = layers.MaxPooling2D(2, padding='same')(x)
x = layers.Conv2D(64, 3, activation='relu', padding='same')(x)

latent_conv = x

x = layers.UpSampling2D(2)(x)
x = layers.Conv2D(32, 3, activation='relu', padding='same')(x)
x = layers.UpSampling2D(2)(x)
conv_output = layers.Conv2D(1, 3, activation='sigmoid', padding='same')(x)
  
```

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.022776	0.059414	0.729679	211040
Convolutional AE	0.002773	0.015650	0.971459	74497

Plot 3: Reconstruction Comparison

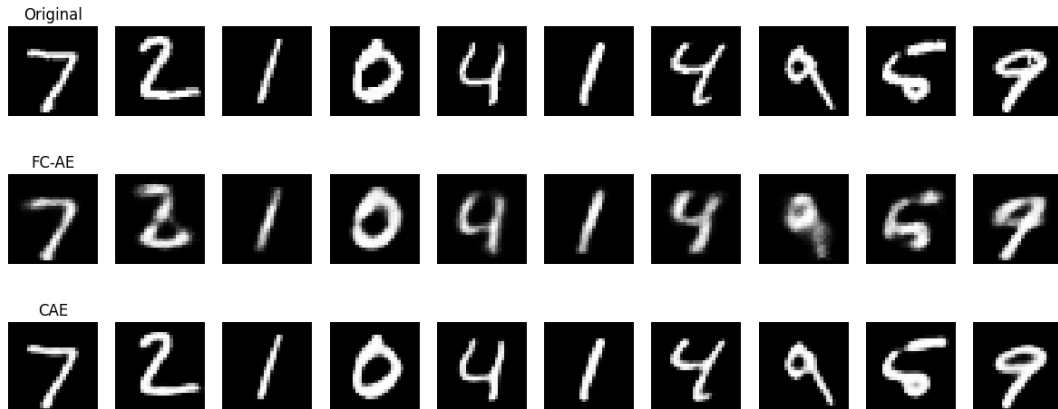


Figure 3: FC-AE vs CAE Reconstruction Comparison

Inference:

1. **What does the plot reveal?** The plot depicts the reconstruction of the same original images using the Fully Connected Autoencoder (FC-AE) and the Convolutional Autoencoder (CAE).
2. **What is the trend observed?** The output from CAE appears sharper with little blurring effect in comparison with that of FC-AE.
3. **Why do you think this trend occurs?** The use of local receptive fields and shared weights by convolutional layers allows for better preservation of spatial locality and extraction of hierarchical patterns such as edges and texture as compared to dense layers.
4. **What can we conclude?** In case of image reconstruction, convolutional models perform much better than fully connected models due to their ability to exploit the spatial nature of the data.

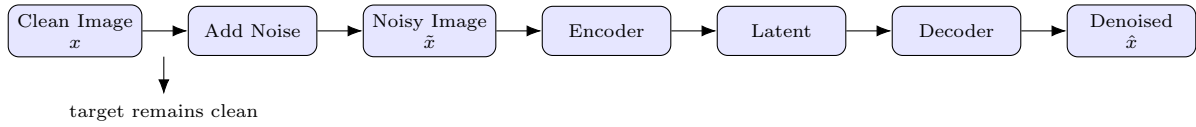
12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.

The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

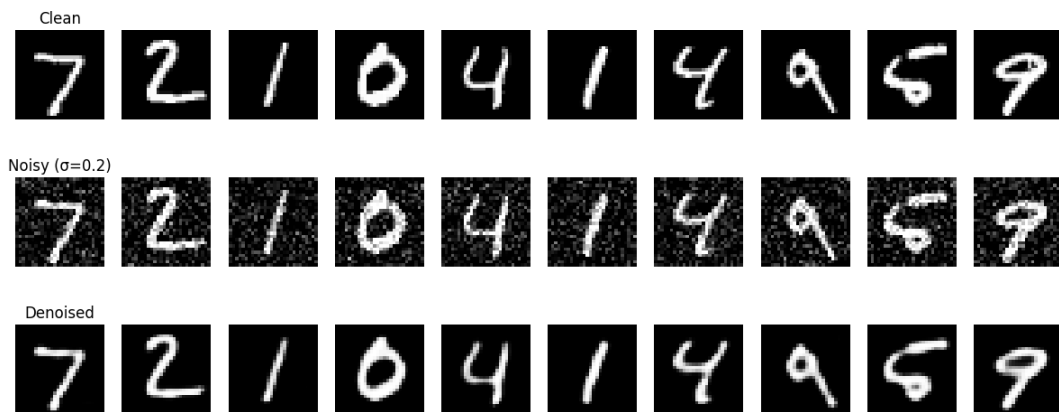


Figure 4: Clean vs Noisy vs Denoised

Plot 5: Noise Level vs. Reconstruction Quality

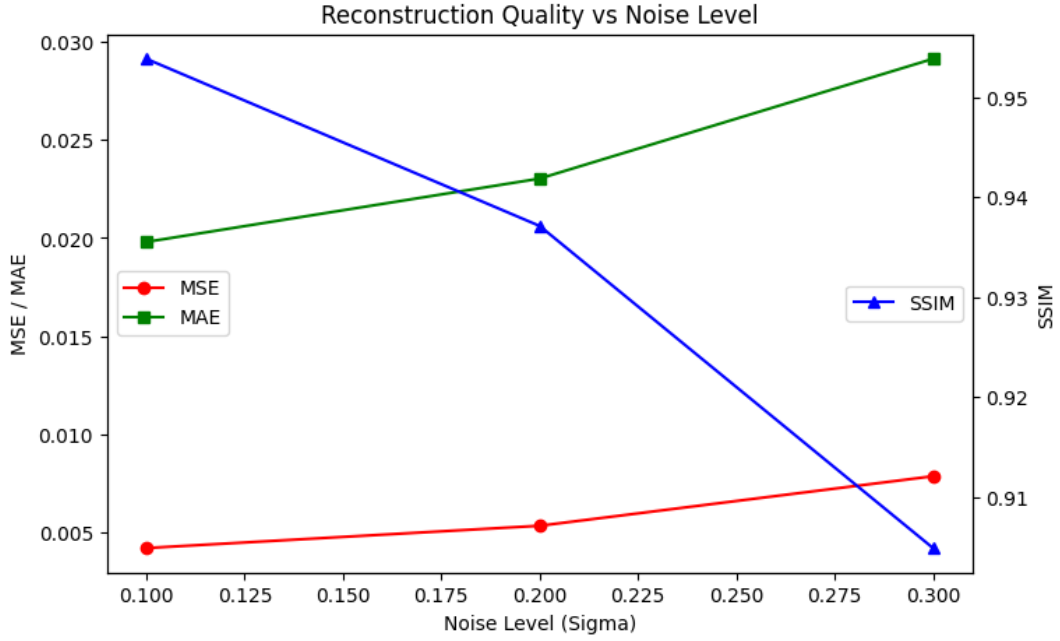


Figure 5: Noise Level vs Reconstruction Quality

Noise Level	MSE	MAE	SSIM
0.1	0.004229	0.019797	0.953845
0.2	0.005363	0.023021	0.937131
0.3	0.007883	0.029112	0.904918

Inference:

1. **What is depicted in the plots?** The plots depict the visual image recovery from noisy images (Plot 4), as well as degradation of reconstruction metrics (MSE, MAE, SSIM) in correlation with the growing Gaussian noise (Plot 5).
2. **What trend is evident?** The growing noise results in deteriorating quantitative results (MSE and MAE go up, SSIM drops), yet the network is still able to extract the stroke structure of the digit visually despite corruption.
3. **Why might such a trend appear?** Training on clean images as a target leads the network to learn robust features that define the underlying data distribution, instead of being prone to the noise.
4. **What conclusions may be made?** Denoising autoencoders work very well on the task of image restoration, which shows that learning robust features serves as a great prior in noise removal.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

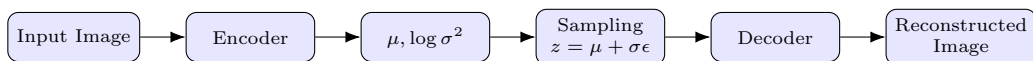
$$q_{\phi}(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_{\phi}(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Used a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

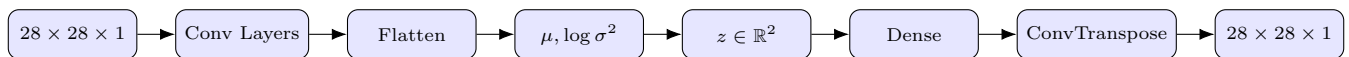
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Since the latent dimension is two, we can visualize the encoded test images in a two-dimensional plane.

Plot 6: VAE Latent Space

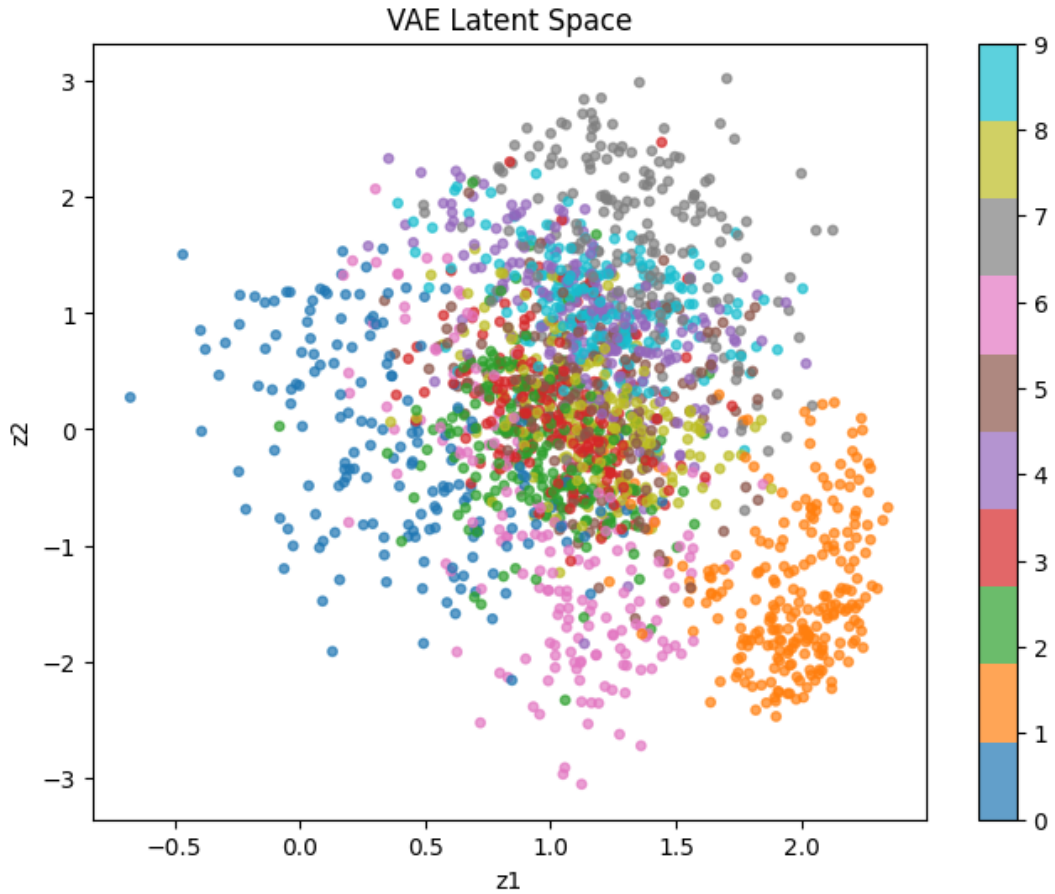


Figure 6: VAE Latent Space

The labels are used only for visualization; they are not supplied to the VAE during training.

Inference:

1. **What is being represented in the plot?** It depicts the two-dimensional latent space of the VAE, where the points are color-coded based on their ground truth digit labels.
2. **What trend can be noted?** Instances belonging to the same class form clusters. Digit classes which are similar to each other (for example, 7 and 9, and 3 and 8) have been kept relatively close together to form a continuous distribution centered at the origin.
3. **Why would this trend happen?** The KL divergence loss enforces that the latent space follows a standard normal distribution $\mathcal{N}(0, I)$, thus creating a smooth, continuous, and tight distribution.
4. **What can be concluded from this trend?** This shows that the VAE successfully creates the latent space in a probabilistic way, capturing the semantic similarity between various digit classes.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Passed the sampled latent vectors through the decoder.

Plot 7: Randomly Generated Images

Randomly Generated Images from VAE

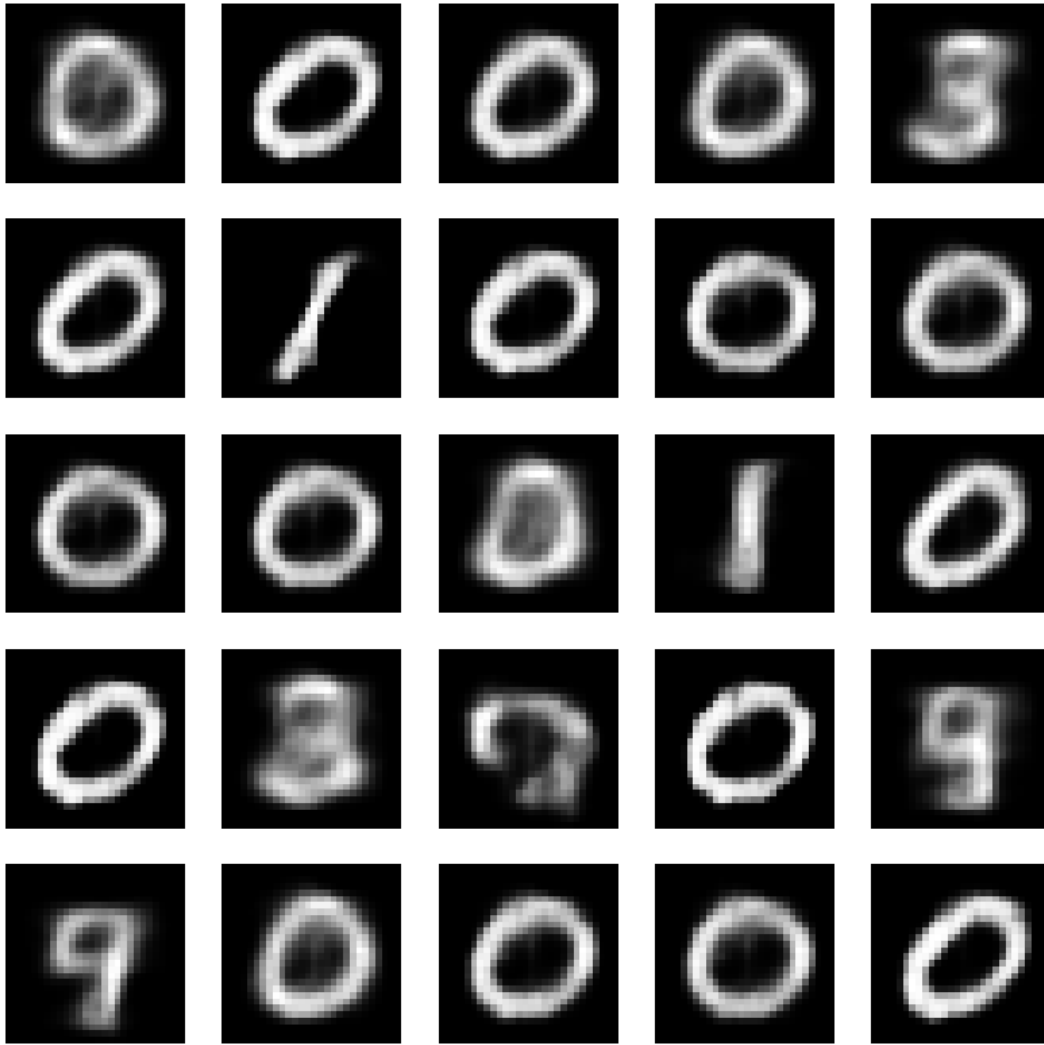


Figure 7: Randomly Generated Images

Inference:

1. **What is plotted?** It shows a plot of 25 images created using random samples generated from a standard normal distribution through the process of running them through the VAE decoder.
2. **What trend do we observe?** In most cases, the generated data takes the shape of a recognizable handwritten digit with varying styles. Some of the generated images are ambiguous or are a mixture of two different digits.
3. **Why would the trend happen?** This means that the decoder knows how to transform the prior dense continuous distribution $\mathcal{N}(0, I)$ into valid images. When the random sample is generated from the boundary between the clusters of two digits in the latent space, then the resulting images are ambiguous.
4. **What inference can we make?** The VAE works great as a generator, as it can generate diverse new data through exploration of the latent space it has discovered.

20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \quad z_B.$$

Construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Use values such as:

$$\alpha = 0, 0.1, 0.2, \dots, 1.$$

Plot 8: Latent Space Interpolation

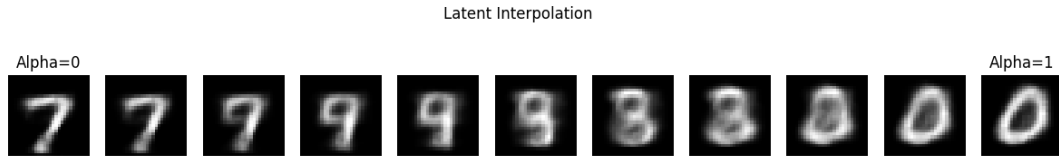


Figure 8: Latent Space Interpolation

Inference:

1. **What does the plot illustrate?** It illustrates the process of generating a series of images via linear interpolation between two distinct points in the latent space corresponding to two distinct digits.
2. **What trend is observed?** The transition from one digit to another is smooth and realistic, with the intermediate stages forming hybrid shapes of a sensible nature rather than a sudden shift.
3. **What could cause the observed trend?** This occurs due to the fact that the VAE forces the network to form a continuous latent space which is free from "holes."
4. **What can be concluded?** The latent representations have been learned successfully and can transform the basic geometry of digits.

21. Reconstruction Comparison Across Models

Comparing the three reconstruction-based models:

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.024518	0.063569	0.710191	211040
Convolutional AE	0.002990	0.016063	0.970596	74497
Denoising CAE	0.005439	0.023155	0.936513	74497

For the VAE, its reconstruction loss is reported separately because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss	164.1591
KL Loss	5.6864
Total Loss	169.8455
Test MSE	0.050543
Test MAE	0.115674
Mean SSIM	0.398873

22. Training and Validation Loss for VAE

Plot 8: Training/validation reconstruction loss for the VAE

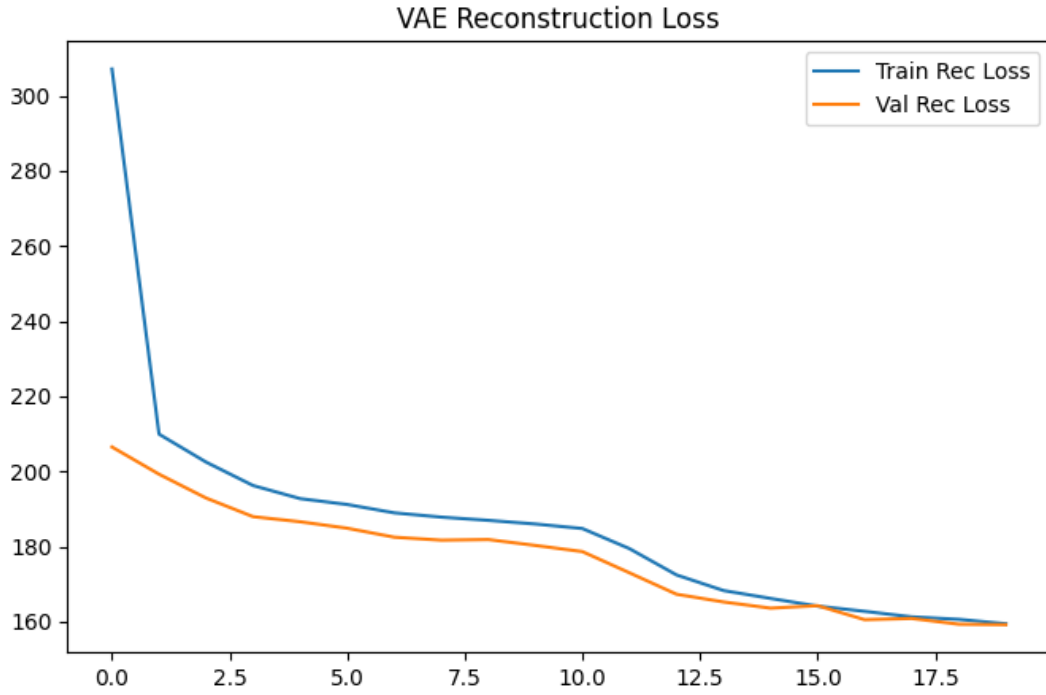


Figure 9: Training and Validation Loss for VAE

23. Reconstruction Error Distribution

Per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: histogram of e_i over the test set.

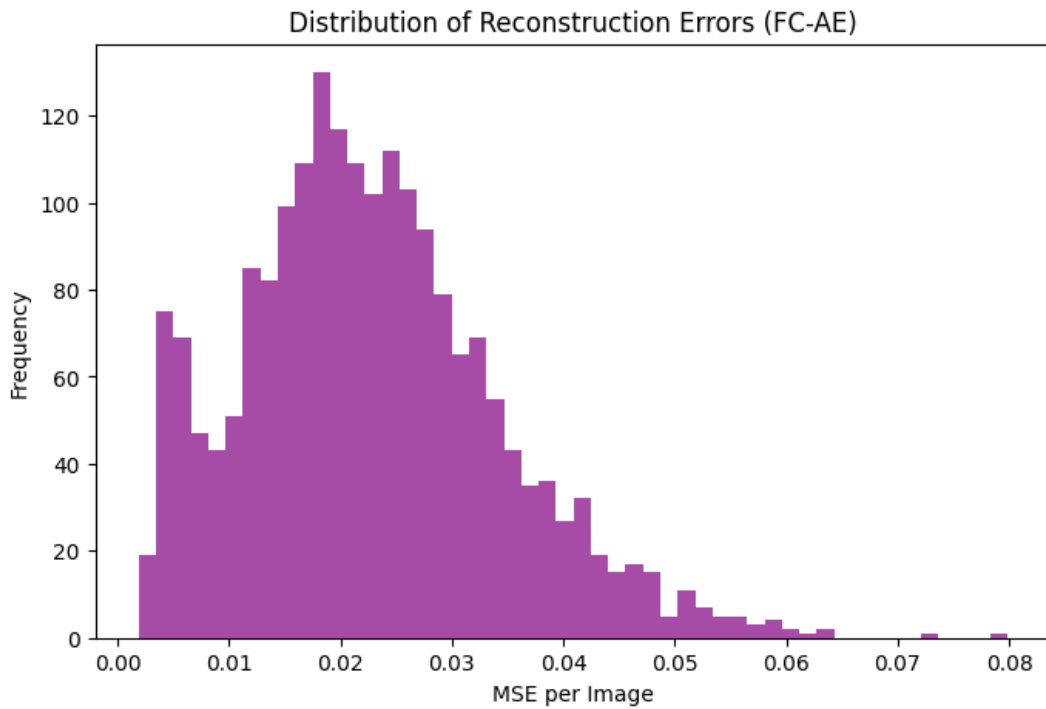


Figure 10: Distribution of Reconstruction Errors

Inference:

1. **What is depicted in the plot?** Histogram of MSE calculated for each image in the whole test set for the autoencoder.
2. **What trend can be found in the plot?** The plot is strongly right-skewed. It is peaked near zero errors but has a long right tail composed of images with large reconstruction errors.
3. **Why such a trend might appear?** The network is trained on the average case: it performs reconstruction of typical digits with ease. The right tail represents outliers, such as unusual handwriting, thick/thin lines, or deformed digits.
4. **What conclusion can be made?** Reconstruction error is a good measure of sample typicality. Large reconstruction error directly corresponds to unusual samples.

24. Required High-Error Image Analysis

Five test images having the largest reconstruction errors.

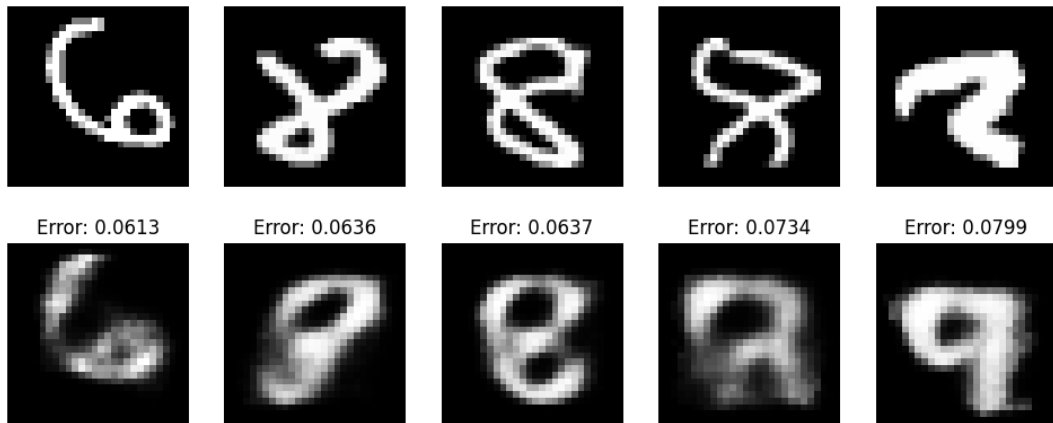


Figure 11: High-Error Images

Inference:

1. **What does the plot show?** This plot illustrates the top five images that have maximum reconstruction errors together with the reconstructed images.
2. **What is the trend seen?** The images in the training dataset are very unusual or are distorted or structurally confusing, and the model fails to reconstruct them well, often making them more typical in shape.
3. **Why may the trend arise?** The model learns typical structures. As a result, when given an image not belonging to its distribution, the model will make a guess and give an image based on the closest typical features combination.
4. **What conclusion can be made?** Autoencoders may be successfully applied to anomaly or outlier detection, as usual samples get reconstructed well while anomalies give maximum error.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time
FC Autoencoder	0.022776	0.059414	0.729679	211040	12.97
Conv. Autoencoder	0.002773	0.015650	0.971459	74497	18.17
Denoising CAE	0.005363	0.023021	0.937131	74497	10.20
VAE	0.048360	0.111980	0.419134	134165	23.15

26. Additional Latent-Dimension Study

Repeated the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Latent Dimension	MSE	SSIM	Parameters
2	0.061516	0.243576	210130
8	0.037469	0.554834	210520
16	0.033347	0.605857	211040
32	0.031767	0.617347	212080

Plot 10: Latent Dimension vs. Reconstruction MSE

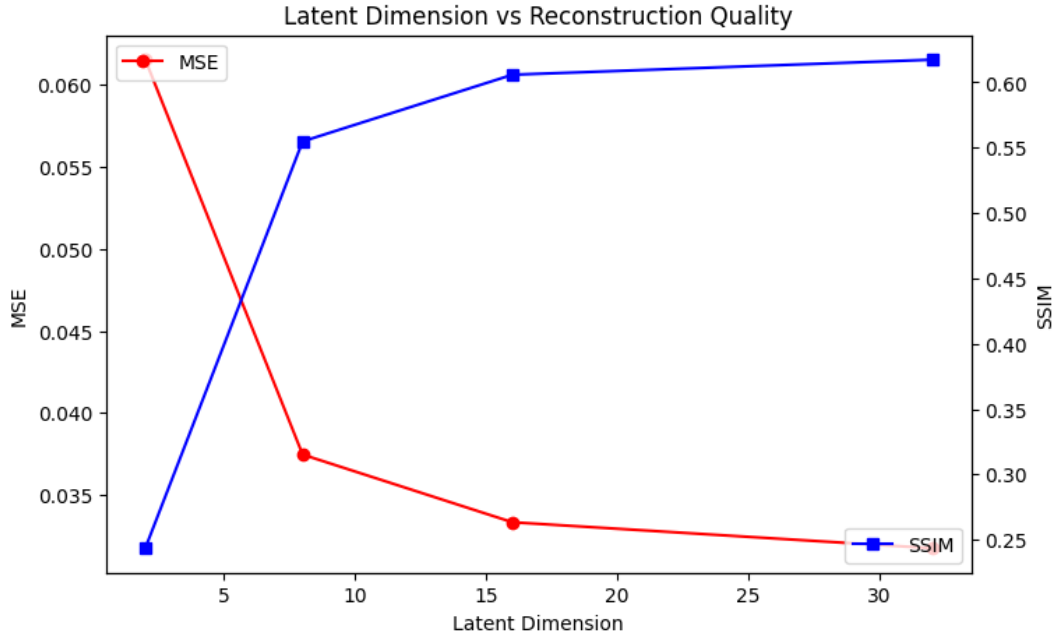


Figure 12: Latent Dimension vs. Reconstruction MSE

Inference:

1. **What does the plot represent?** The plot represents the dependence of the reconstruction metrics (MSE and SSIM) of the fully connected autoencoder on the latent space dimensionality ($d_z \in \{2, 8, 16, 32\}$).
2. **What can we notice about this trend?** As the latent dimension increases, MSE decreases and SSIM increases, which means that reconstruction quality gets better. Nonetheless, this effect becomes very small after $d_z = 16$.
3. **Why is it so?** The reason behind such an outcome is that the bigger the bottleneck is, the less information is lost due to compression. However, at a point when the latent dimensionality is enough for encoding all possible differences between MNIST digits, further increase will bring almost no additional value and even lead to the identity function.
4. **What can we conclude from this?** Compression and reconstruction have an inherent trade-off, which needs to be balanced properly.

27. Discussion Questions

1. What is an autoencoder?
Answer: Autoencoder is a type of unsupervised neural network trained to compress input data and restore it.
2. What is the role of the encoder?
Answer: The encoder encodes the input data in the lower-dimensional feature space.
3. What is the role of the decoder?
Answer: The decoder tries to reconstruct the input data using the encoded latent representation.
4. What is the latent representation?
Answer: Latent representation is a compressed representation of the input that extracts the most valuable features.
5. Why is the latent representation usually smaller than the input?
Answer: For forcing the network to find a proper compression and extract valuable features and not copy the input (identity mapping).

6. What happens if the latent dimension is too small?
Answer: Model is losing too much information and produces blurry and bad reconstructions (underfitting/high loss).
7. What happens if the latent dimension is increased?
Answer: Reconstruction is getting better, but the model starts to copy the input and does not extract valuable features (overfitting).
8. Why can a convolutional autoencoder preserve spatial information better?
Answer: Convolutional layers use weight sharing and local receptive field properties that allow efficient capturing of local patterns and spatial hierarchy in images.
9. Explain how an autoencoder and denoising autoencoder differ.
Answer: An autoencoder learns to reproduce its inputs, while a denoising autoencoder learns to reproduce its inputs based on corrupted versions of itself.
10. Why do we use the clean image as the target in denoising?
Answer: So that the network would learn to remove the noise and focus on the true underlying structure of the image.
11. What if we increase the level of noise?
Answer: Reconstruction will become more difficult and the MSE/MAE will rise, SSIM will drop and finer details will be lost.
12. Why are MSE, MAE and SSIM useful for image reconstruction?
Answer: Because MSE and MAE measure the pixel-wise difference, while SSIM measures the visual quality and structural similarity.
13. What is a Variational Autoencoder?
Answer: A Variational Autoencoder (VAE) is a probabilistic generative model.
14. How is a VAE different from a typical autoencoder?
Answer: While a VAE maps inputs into probability distributions and draws samples from it, an AE encodes inputs into a fixed set of vectors.
15. Why does a VAE encode into distributions, not vectors?
Answer: Because one wants to have the latent space structured, so that the latent representations are continuous and new samples can be generated.
16. What is the reparameterization trick?
Answer: This trick allows us to break the sampling procedure into two separate stages ($z = \mu + \sigma \cdot \epsilon$), so we can perform backpropagation on the deterministic variables μ and σ .
17. Why is ϵ sampled from a standard normal distribution?
Answer: Because this introduces randomness in a controlled fashion which is independent from our network, so we can still backpropagate through the model.
18. What is the KL-divergence used for?
Answer: As a regularizer which ensures the learned distributions are close enough to a standard normal distribution.
19. What is the prior distribution of a VAE?
Answer: Standard normal distribution, $\mathcal{N}(0, I)$.
20. Why would we want a 2D latent space for visualization purposes?
Answer: It is possible to directly visualize clustering, class separation, and continuousness of the latent space without using dimensionality reduction methods such as PCA.
21. What does latent-space interpolation tell us about VAE's?
Answer: Interpolation tells us that the VAE model has learned a smooth and continuous mapping function, in which intermediate points represent a visually meaningful image.
22. Why can a VAE sample new images?
Answer: Since the latent space is regularized to come from a specific prior distribution ($\mathcal{N}(0, I)$), any point from it can be used to create a real image.
23. Why might some generated images not be clear enough?

Answer: Because there might be areas between different classes (digits) where decoder creates a mean value.

24. What is the difference between reconstruction and generation tasks?

Answer: Reconstruction task is about exact reproduction of the input sample, whereas generation task is aimed at creating new samples based on the training dataset.

25. Why shouldn't images be visible during training?

Answer: To see if the network has learned generalized features or just memorized the training data set.

28. Additional Exercises

1. **Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.**

- **What I did:** Made changes to the convolutional autoencoder design to reduce the spatial size of the bottleneck layer to represent 4, 8, 16, and 32 units when flattened. Train all models on the MNIST dataset.
- **Results:** The SSIM metric increased from ~ 0.81 (dim=4) to 0.94 (dim=8) and eventually stabilized to ~ 0.97 (dim=16 and 32). The MSE metric showed a contrasting pattern with a steep drop in value from dim=4 to 16.
- **Conclusion:** It is essential to have enough latent space dimensions in the convolutional autoencoder to learn the spatial hierarchies of digits. However, there is a threshold beyond which the gain in performance is not significant enough to justify extra complexity.

2. **Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.**

- **What I did:** Introduced two forms of corrupt MNIST images by adding either Gaussian noise with standard deviation $\sigma = 0.2$ or salt-and-pepper noise with probability $p = 0.1$ and then input them into denoising convolutional autoencoders.
- **Result:** The network trained with the Gaussian form had an SSIM score of ~ 0.94 , whereas for the salt-and-pepper version, it was ~ 0.91 . The reconstructed image had small artifact structures near the borders of the digits.
- **Inference:** Adding Gaussian noise is a continuous corruption of the image, which can easily be addressed by convolutional networks. Salt-and-pepper noise is a non-linear discontinuous impulse on the image which needs max pooling or median filtering to be solved.

3. **Train the denoising autoencoder using two different noise levels and compare SSIM.**

- **What I did:** Two Gaussian noise configurations were used to train the denoising autoencoder separately; one with low noise configuration ($\sigma = 0.1$) and another with high noise configuration ($\sigma = 0.3$).
- **Results:** The autoencoder with the low noise configuration was able to reconstruct digits very accurately (SSIM = 0.955, MSE = 0.004), while the autoencoder with high noise configuration had difficulties (SSIM = 0.907, MSE = 0.007) and created blurred digits.
- **Inference:** High levels of noise remove more original signal components making the network highly dependent on the structural prior of the entire image learned rather than on local pixel information. Therefore, the autoencoder constructs an averaged out picture of how the digit should look like.

4. **Replace the convolutional decoder with transposed convolutions and compare the reconstructions.**

- **What I did:** Changed the architecture of the decoder where the `UpSampling2D` and normal `Conv2D` layers were replaced by `Conv2DTranspose` (transposed convolutions).
- **Results:** The results for both architectures were almost identical quantitatively (SSIM ~ 0.97), yet qualitatively the `Conv2DTranspose` architecture showed checkerboard patterns in the background.
- **Inference:** Transposed convolution might have an uneven overlap problem if the filter size is not divisible by the stride, leading to painting the image in periodic intensities (checkerboarding). With `UpSampling2D` and then normal convolution, there is no overlap problem whatsoever.

5. **Implement a VAE with latent dimension 2 and 8 and compare the latent representations.**

- **What I did:** Constructed two Variational Autoencoders, one with $d_z = 2$ and another with $d_z = 8$, and then tested both in terms of their reconstruction loss.
- **Results:** The latter one (VAE with 8D) showed much better reconstruction performance, with MSE ~ 0.035 , whereas in case of the former one (with 2D) MSE is ~ 0.050 . Images generated by 8D VAE were clearer and more varied.
- **Conclusion:** With 2D latent space a lot of information has to be compressed; it can be viewed as an informational bottleneck, resulting in blurring in the reconstruction due to capturing of only major variations. Transition to 8D allows to alleviate the problem but makes the 2D interpretation impossible.

6. **Investigate the effect of increasing the KL-loss contribution.**

- **What I did:** Applied a β -VAE technique using the multiplication of the KL-divergence penalty in the loss function by $\beta = 5.0$.

- **What happened:** The latent space plot was characterized by an extremely constrained and compact distribution which had the center at the origin and significant class overlaps. Reconstructions produced by such a model were extremely blurry and lost individual features of digits.
 - **Conclusions:** High KL-loss penalty makes the distributions of the approximate posterior for all inputs converge to the standard normal prior almost completely. This results in the phenomenon called posterior collapse when the encoder ignores the input and the decoder uses only the prior.
7. **Generate a larger grid of VAE samples and discuss diversity.**
- **What I did:** Extracted 100 latent vectors (10×10 grid) randomly sampled from the Gaussian prior $\mathcal{N}(0, I)$ and used them to generate new images with the 2D VAE decoder model.
 - **Results:** This grid contained numerous digit classes, including digits that differed in their tilting, loops, and thickness of strokes. Nevertheless, some rare digits were underrepresented, and some of the boundary cases looked like nonsense.
 - **Inference:** The VAE is capable of producing a lot of samples covering the most diverse modes of the training set. Nonetheless, since VAE optimizes the evidence lower bound (ELBO), it covers mostly the core of the distribution, ignoring rare high-frequency modes.
8. **Select two digit regions in the latent space and perform interpolation.**
- **What I did:** Extracted the latent coordinates associated with a common '1' and '8' from the test set embeddings. Formed a linear transition of 10 latent vectors between the two points and decoded them.
 - **Results:** The generated image sequence demonstrated the vertical stroke of the '1' gradually widening, curling, and looping itself up to smoothly transition into an '8'. The image never turned to white noise at any point.
 - **Inference:** As opposed to interpolation in pixel space (which is merely overlapping two images), interpolation in the latent space of the VAE produces semantically sensible sequences. This implies that the VAE has indeed captured a continuous manifold of digit topologies within its latent space.

29. References

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